Detecting Emerging Patterns in Bank Card Fraud Using a Neuroadaptive Deep Learning Framework

Babatope Makinde
Centre of Intelligence of Things
University of Greater Manchester
Bolton, England
babatope.makinde@ieee.com
0009-0000-4829-1714

Celestine Iwendi
Centre of Intelligence of Things
University of Greater Manchester
Bolton, England
c.iwendi@bolton.ac.uk
0000-0003-4350-3911

Olayinka Anthony Ojo
Centre of Intelligence of Things
University of Greater Manchester
Bolton, England
A.Ojo@greatermanchester.ac.uk

Salome Enoshi Uwah
Centre of Intelligence of Things
University of Greater Manchester
Bolton, England
s.uwah@ieee.org
0009-0004-2288-2654

Tosin Fadeyi
Centre of Intelligence of Things
University of Greater Manchester
Bolton, England
tf1res@bolton.ac.uk
0009-0006-1322-8583

Collins Lemeke
Centre of Intelligence of Things
University of Greater Manchester
Bolton, England
collinslemeke@ieee.org
O0009-0001-3192-7461

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***Abstract*—Bank card fraud is one of the biggest challenges in digital finance space, which needs detection models to address class imbalance, interpretability, and adaptability to changing tactics of fraud. The paper proposes a neuro-adaptive architecture established on a highly structured preprocessing pipeline with stratified splitting, feature normalisation, and representation learning via a Denoising Autoencoder. At the core of this framework lays an Artificial Neural Network optimised by the Firefly Algorithm for fast hyperparameter tuning facilitated by Elastic Weight Consolidation that promotes continual learning without sacrificing past performance. The proposed Adaptive ANN + FA outperforms baseline ANN, CNN, and LSTM models mainly in F1-Score, precision, and recall-the main metrics in fraud detection. Also, SHAP breaks out feature contribution and prediction reasonability making the results very transparent. Optimised adaptive and explainable models are positioned here as strong enablers of real-world fraud discovery and subsequent robustness in the financial systems.**

***Index Terms*—Fraud detection, Payment Cards, Machine Learning, Deep Learning, Neuroadaptive, Continual Learning, Firefly Algorithm, Optimisation, Sampling, Hyperparameter.**

I. Introduction

Payment card fraud is a growing concern worldwide, exacerbated by the rapid expansion of digital payments, e-commerce, and contactless transactions [1]. This study will develop a neuro-adaptive framework to detect the emerging pattern in bank card fraud. This report contains the design of a neuro-adaptive model to detect emerging pattern in bank card fraud, performance and evaluation of the proposed model.

A. Background

The digitalisation of financial services has brought about easy use of payment card systems where losses start accumulating from the point convenience is accompanied by high vulnerability to such advanced threats as phishing, identity theft, and card skimming. Losses have grown to more than$35 billion in 2022 recorded globally [2] and approximately £1.2 billion in the UK during 2023 [3]. See Figures 2 and 3. Conventional protection schemes against fraud that are predominantly rule-based or signature-based cannot provide dynamic protection against continuously evolving approaches to fraud. Since there is no certain pattern to malicious actions, static models soon become obsolete. This, therefore, sets up the imperative challenge for adaptive detection methodologies to learn from new transactional data and dynamically adjust for emerging threats.

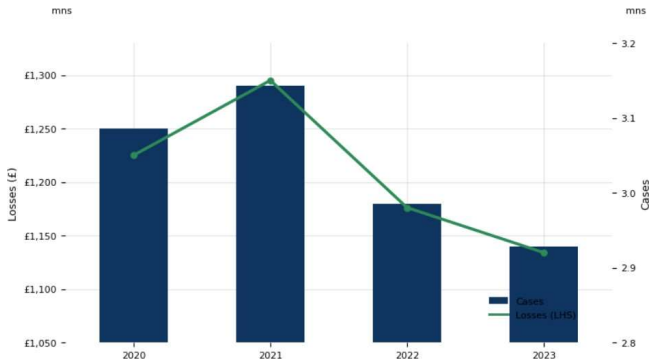

Figure 2. Fraud losses and voluming of cases (UK Finance, 2024)

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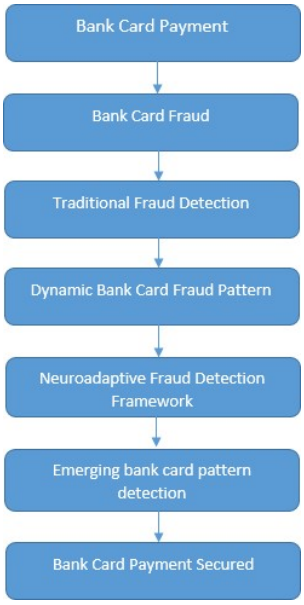

Figure 1. System Architecture, Author's

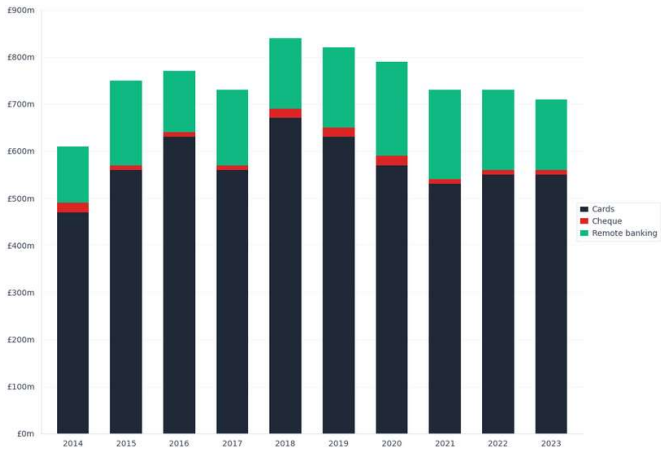

Figure 3. Annual Value of Fraud Losses by Payment Method in the UK (UK Finance, 2024)

B. Problem Statement

A constant global menace when accompanied by massive annual losses, is what bank card fraud can best be described as. The static rule-based or machine learning models on which the current systems depend fail to identify changing strategies of fraud effectively. As soon as a single method is recognized and utilised by fraudsters, it gets outdated and does not cater to the dynamically formed fraudulent behaviors.

C. Aim of Study

The aim of the study is to develop a Neuro-adaptive model for bank card fraud detection that can intelligently spot changes, learn from the new patterns and fix class imbalance in bank card fraud.

D. Research Objectives

- To review and critically analyse existing bank card fraud detection techniques with a focus on their limitations in handling evolving fraud patterns, data imbalance, and real-time adaptability.
- To acquire and preprocess a large real-world bank card transaction dataset, making sure fraud and legitimate cases are balanced for the maximum effectiveness in training the model.
- To design and implement fraud detection framework with an emphasis on adaptivity to learn and detect new patterns in bank card frauds.
- To apply a nature-inspired metaheuristic algorithm to optimise the hyperparameter of the network to improve the adaptiveness and detection accuracy..
- To implement the adaptive model and test it using metrics, compare with baseline model and ensure interpretability.

E. Significance of Study

The project puts its focus on improving fraud detection models in terms of real-time adaptability and predictive accuracy as a way or means for financial institutions to enhance cybersecurity. This proposed solution shall ensure scalability, reduced operations costs, and enhanced customer confidence amidst changing trends in digital finance and emerging patterns of fraud.

II. Related Works

This review takes a hard look at thirteen recent studies, laying out their goals, importance, and methods while putting them into groups based on common approach, standout results, fresh ideas, and different takes on how things should be done by look another way.

A. Unified Goals and Shared DL Techniques in Transaction Fraud Detection

Many of the studies reviewed here take related methodological approaches, i.e., an approach based on deep learning grounded in convolutional neural networks (CNNs) for detecting fraud. [6], [7] and [8] Assessing CNN's Performance with Multiple Optimization Functions for Credit Card Fraud Detection, all build classic CNN architectures to learn discriminative features directly from transaction records. Just as [9] begins with a feature extraction stage based on CNN, it extends further by including RNN components. [10] focuses on benchmarking a range of conventional machine learning models against deep neural networks and demonstrates that neural-based models consistently outperform classical algorithms on financial transaction datasets.

B. Standout Contributions and Outstanding Architectures that Redefine Fraud Detection Models

A few works shine either through great results or by their tough test plans. The paper by [11] stands out since it deals with class imbalance; one of the most pressing issues in fraud spotting using a well-picked team of deep learners whose weights are shifted as needed over time. Also, the work by

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[12] sets itself by adding many strong base learners or by careful work on transaction micro-features that bring fraud recall to the maximum level without raising the number of false positives. [13] puts forward an architecture optimised remarkably with weightless layers together with a well-tuned learning rate schedule.

C. Innovative Approaches and Emerging Perspectives in Fraud Detection

In term of novelty, some papers lead with new research directions beyond the typical CNN/RNN frameworks. [14] proposed a meta-learning framework which learns from very few examples of fraud, because learning from ever-evolving scenarios is most relevant. [15] presents a federated learning system in which models trained at separate institutions perform adaptive model fusion to deal with heterogeneous feature spaces. [16] takes the art one step further by merging concept drift detection with retraining triggers that are urged adaptively. [17] applies reinforcement learning in combination with an LSTM network, which dynamically adjusts its decision policy in the presence of novel transaction patterns.

D. Diverse Approaches and Varying Depths

The reviewed papers also differ in deployment environment, preprocessing strategies, and integration into the system. [18] architectural design and cloud deployment optimization were more emphasized rather than presenting a new algorithmic model. Such other works as [9] present how time-series dependencies can be exploited apart from spatial patterns that have always been considered. Some describe more of the data preprocessing stages-drift detection, adaptive windows, feature engineering, some discuss more of the operation aspects of data privacy and computational scalability-federated learning and cloud-based deployment.

III. METHODOLOGY

This chapter discusses the methodology used in exploring and developing a strong, understandable, and flexible structure for catching emerging fraud in card transactions. The research setup is an actual study, using a multi-step process that brings together data science, machine learning, and explainable AI methods to tackle issues of class imbalance, model effectiveness, and shifts in concepts usually found in financial data flows.

Figure 4. Work Flow Procedure Chart, Author's

A. Research Philosophy and Design

This study falls under the positivist research philosophy. That particular philosophy places emphasis on objectivity, measurable evidence, and a systematic model assessment through empirical testing. Thus, this study uses a quantitative structured approach to design and implement a neuroadaptive deep learning framework for bank card fraud detection. To ensure rigorous evaluation as well as contribute toward adaptability plus interpretability improvement in fraud detection systems, it applies preprocessing and feature engineering plus optimization pipelined systematically benchmarked against existing state-of-art deep learning models. See Figure 6 below.

B. Proposed Neuroadaptive Framework

The proposed bank card fraud detection system uses deep learning to overcome the inadequacies of traditional machine learning discussed in section IV by starting with data preprocessing and scaling, stratified splitting into training, validation, and test sets then apply class balancing techniques such as SMOTE, SMOTE-ENN, and ADASYN to the training set only. Model development will be carried out using 5-fold cross-validation where hyperparameters are optimized by the Firefly algorithm resulting in an adaptive ANN architecture enhanced with PReLU activation, Batch Normalization plus AdaBN as well as Elastic Weight Consolidation for continual learning. Training includes early stopping and check-pointing while SHAP makes it interpretable; finally benchmarking the fully optimized model against a baseline to prove its better performance as described in section 1.6 and Figure 4.

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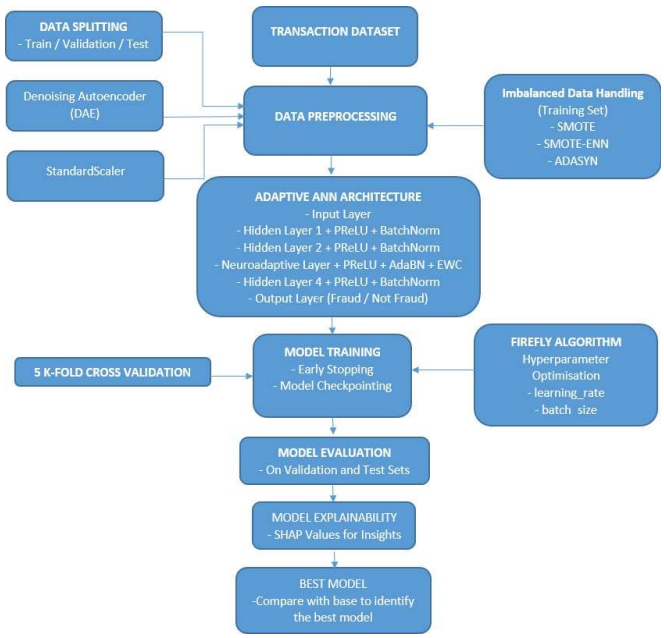

Figure 5. Proposed Neuroadaptive Framework, Author's

C. Data Collection and Preprocessing

The study utilizes a publicly available (Kaggle), anonymized dataset of bank card transactions. The dataset has a high level of imbalance where the fraudulent transactions make up less than 10 percent of all the records as described in section 4.6.1.

Data preprocessing, scaling, and sampling are also included a part of the guarantee for stability and fairness of the model. At input level, StandardScaler has been applied by subtracting the mean and dividing by standard deviation (= 0, = 1) to help stabilize the model [10]. The stratified data split preserves class distribution in train and test splits. This is considered best practice in machine learning. Oversampling of minority classes will be used for resolving the unbalanced problem between majority classes using SMOTE which may introduce noise [13], SMOTE ENN which removes noise produced while oversampling [12], ADASYN which creates samples adaptively in regions that are difficult to learn [16]. Denoising autoencoders (DAE) will be used for reconstructing clean inputs from noisy data hence highlighting true patterns for training and learning of models [11] through activation functions leading to accuracy enhancements by shrinking negative values hence avoiding dead neurons [19]. Batch Normalisation (BN) normalizes activations [9] while Adaptive Batch Normalisation (AdaBN) replaces the BN statistics with the statistics computed from the target domain during inference thereby accounting for any domain shift without involving retraining of the network [20]. The Firefly Algorithm, a metaheuristic that is inspired by and based on the flashing behavior of fireflies, has been applied to search for optimal solutions in parameter space; thus improving accuracy, fairness, and adaptivity [21]. Also, Elastic Weight Consolidation (EWC) preserves important weights with the help of Fisher information against catastrophic forgetting throughout new tasks[22]. Training robustness was facilitated by early stopping that would not allow overfitting[12], checkpointing saving and restarting efficiently the model states as well as 5-fold cross-validation that reduces overfitting and strengthens model generalization[7].

D. Model Comparison Framework

To evaluate the efficacy of the proposal model, it is benchmarked against three baseline deep learning models LSTM, CNN, and ANN.

E. Training, Validation and Testing

Mathematically, the proposed Neuroadaptive ANN structure is represented as follows:

Input layer: $x = [x_1, x_2, \ldots, x_n]$

Hidden layer 1: $h_1 = \mathrm{BN\ PReLU}(W_1 x + b_1)$

Hidden layer 2: $h_2 = \mathrm{BN\ PReLU}(W_2 h_1 + b_2)$

Hidden layer 3 (Neuroadaptive): $z_3 = W_3 h_2 + b_3$, $a_3 = \mathrm{PReLU}(z_3)$, $a_3' = \mathrm{AdaBN}(a_3)$, $h_3 = \mathrm{EWC}(a_3')$

Hidden layer 4: $h_4 = \mathrm{BN\ PReLU}(W_4 h_3 + b_4)$

Output layer: $z_0 = W_0 h_4 + b_0$, $\sigma(z_0) = \frac{1}{1+e^{-z_0}}$.

x: input feature vector; W_i: weight matrix for layer i; b_i: bias vector for layer i; σ: sigmoid activation function [23]

IV. Results

During the experimental phase, each benchmark model (LSTM, CNN, ANN and Proposed Neuroadaptive Model) are trained on 70 percent of the dataset and then evaluated on the remaining 30 percent (15 percent validation and 15 percent test) to objectively assess predictive performance on previously unseen transactions

Figure 6. Loss Graph, Python Output

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Figure 7. Accuracy Graph, Python Output

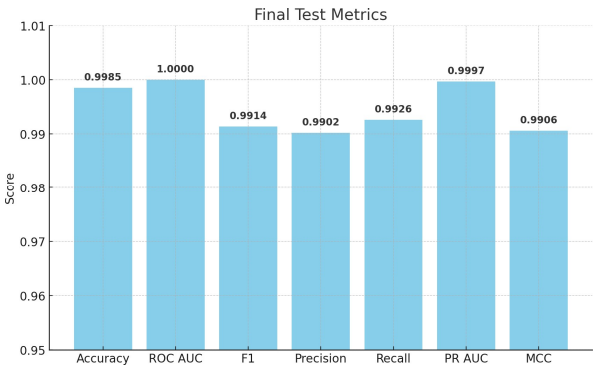

Figure 8. Final Testing Metrics, Python Output

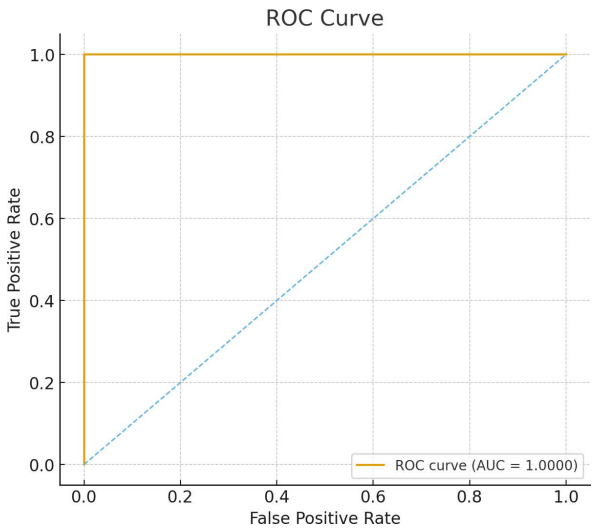

Figure 9. Roc Curve, Python Output

At the evaluation stage, the LSTM, CNN, and ANN models join the Proposed Neuroadaptive model in being run on a held-out test set. Standard metrics for performance Accuracy, Precision, Recall, F1-Score is accompanied by a confusion matrix to allow an objective comparison of how well these models can identify fraudulent transactions.

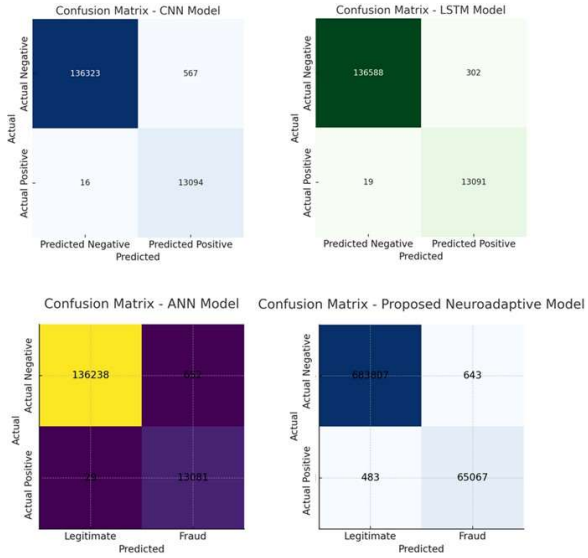

Figure 10. Confusion Matrix of experimented models, Python Output

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS, PYTHON OUTPUT

Models	Accuracy	Precision	Recall	F1-Score	AUC-ROC
ANN+FA	99.85%	99.02%	99.26%	99.14%	100%
LSTM	99.79%	97.75%	99.86%	98.79%	99.99%
CNN	99.61%	95.85%	99.89%	97.82%	99.99%
ANN	99.56%	95.25%	99.78%	97.46%	99.99%

V. DISCUSSION

The Proposed Neuroadaptive ANN + FA attains an Accuracy of 99.85%, Precision of 99.02%, F1-Score of 99.14% and a Recall of 99.26%, thus proving the effectiveness of the design. Reflectively from section 1.4.1, stating the research objectives. The achievement of proposed model is observed by its ability to performing better than all baseline models as shown in Table 5 and Figure 8. The framework's strength lies in its continual learning and interpretability, making it a robust and practical solution for bank card fraud detection.

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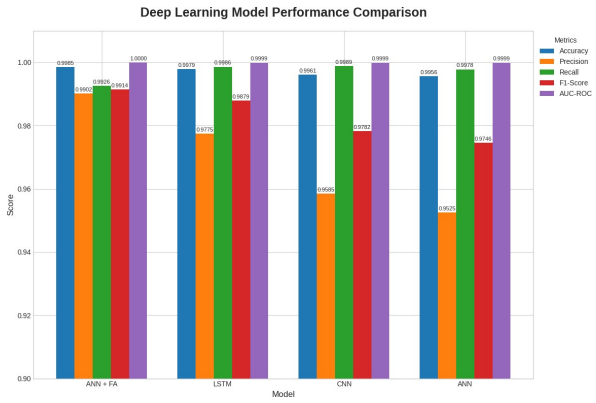

Figure 11. Model Comparison, Python Output

The SHAP analysis picks this feature as the top one since its values always have the strongest effect on the model's output, clearly separating fraud from legit transactions as described in Figure 9.

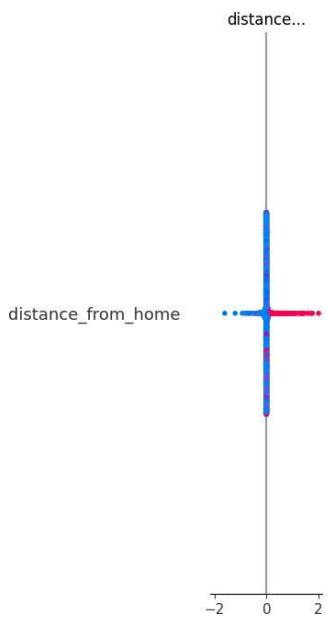

Figure 12. SHAP Visualization, Python Output

VI. Conclusion and Future Work

This study presents a novel framework for bank card fraud detection, designed to overcome the critical limitations of existing models. By integrating a multi-stage approach and encompassing intelligent feature engineering, hyperparameter optimisation, continual learning. This research successfully developed a robust, transparent, and adaptive solution. This research addressed three primary questions, and the findings confirm the viability and efficacy of the proposed framework.

Can an adaptive machine learning model be developed for bank card fraud? Yes, the neuroadaptive framework applies Elastic Weight Consolidation (EWC) thereby allowing the model to be continuously fine-tuned on new unknown data as a way of effectively responding to concept drift by learning new patterns of fraud without unlearning old ones.

What are the challenges faced by previous study of bank fraud detection? The study addressed three major challenges: class imbalance with stratified splitting and sampling techniques, "black box" models with explainability through SHAP, and concept drift with a continual learning strategy.

Can a Neuro-Adaptive Machine Learning Model be better than base line model? YES: The Neuro-Adaptive framework leveraging Denoising Autoencoder, Firefly Algorithm and neuroadaptive mechanisms beats the simple models by achieving a higher Accuracy Precision, F1-score and Recall which are the most important metrics in fraud detection.

This framework leads the industry and academicians toward the development of robust fraud detection systems. It is an end-to-end pipeline covering all aspects of the system lifecycle scope, from data processing up to adaptability. Future works need to use it as a baseline, filling its gaps in limitations and bringing new architectures and optimization strategies.

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